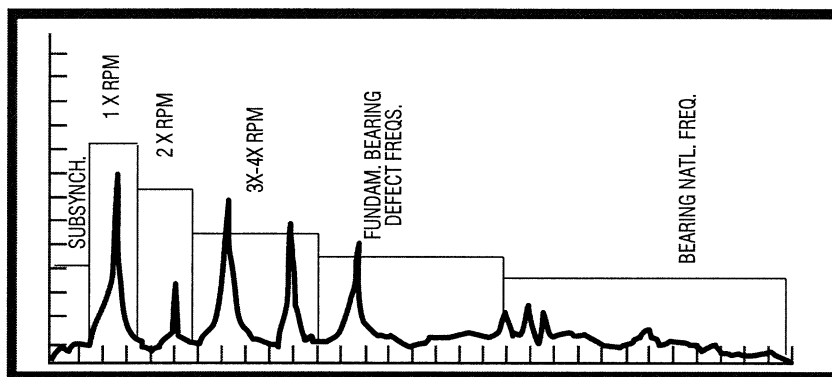




ANALYSIS I

How To Implement An Effective Condition Monitoring Program Using Vibration Analysis



Authored By:

Mr. James E. Berry P.E.

of Technical Associates of Charlotte, P.C.

Technical Associates Of Charlotte, P.C.

347 North Caswell Road Charlotte, NC 28284, U.S.A. TELEPHONE: (704) 333-9011 FAX: (704) 333-1728

www.technicalassociates.net

SPECIALISTS IN PREDICTIVE MAINTENANCE, MACHINERY DIAGNOSTICS, AND VIBRATION REDUCTION

PREDICTIVE MAINTENANCE AND VIBRATION SIGNATURE ANALYSIS I

By: James E. Berry, P.E., Mechanical Engineer

Technical Associates of Charlotte, P.C.

347 N. Caswell Rd.
Charlotte, NC 28204
(704) 333-9011

**SPECIALISTS IN PREDICTIVE MAINTENANCE, MACHINERY DIAGNOSTICS,
AND VIBRATION REDUCTION**

We thank you for the opportunity to assist you in development of your vibration analysis and condition monitoring skills. As you can see from the length and breadth of the information within this text, many variables go into application of the material and so much depends upon the knowledge and judgement of the analyst and/or technician. The author and his company and /or companies can assume no responsibility for results in spite of many years of successful and trouble-free personal application.

First Edition: February, 1994

Second Edition: June, 1997

Copyright © 1990, 1991, 1992, 1993, 1994 & 1997 Technical Associates of Charlotte, P.C.
All rights reserved. This booklet, or parts thereof, may not be reproduced in any form without written permission from an officer of the company.

TABLE OF CONTENTS AND SEMINAR AGENDA

PREDICTIVE MAINTENANCE AND VIBRATION SIGNATURE ANALYSIS I

<u>SECTION</u>	<u>SUBJECT</u>	<u>PAGE</u>
1.	ANALYSIS I SEMINAR OVERVIEW.....	1-1
2.	WHAT IS VIBRATION AND HOW CAN IT BE USED TO EVALUATE MACHINERY CONDITION?.....	2-1
2.0	Introduction.....	2-1
2.01	What is Vibration Frequency and How Does it Relate to a Time Waveform?.....	2-3
2.02	What is Vibration Displacement?.....	2-3
2.03	What is Vibration Velocity?.....	2-4
2.04	What is Vibration Acceleration?.....	2-4
2.05	What is Vibration Phase?.....	2-5
2.1	What is a Vibration Spectrum (Also Called an "FFT" or "Signature")?.....	2-7
2.2	Difference Between RMS, Peak and Peak-To-Peak Vibration Amplitude?.....	2-11
2.3	When to Use Displacement, Velocity, or Acceleration?.....	2-13
2.4	How Much is Too Much Vibration?.....	2-14
2.5	Understanding a Vibration Spectrum.....	2-22
2.51	Effect on Frequency Accuracy of #FFT Lines Used.....	2-22
2.52	Effect on Frequency Accuracy of Frequency Span Used.....	2-24
2.6	What is Overall Vibration (Digital and Analog Overall Level)?.....	2-33
2.7	Understanding Phase and Its Applications.....	2-37
2.71	Definition of Phase.....	2-37
2.72	How to Make Phase Measurements.....	2-37
2.73	Using Phase Analysis in Vibration Diagnostics.....	2-39
2.731	Evaluating Axial Motion of a Bearing Housing to Reveal a Possible Cocked Bearing or Bent Shaft.....	2-39
2.732	Phase Behavior Due to Unbalance.....	2-40
2.733	Phase Behavior Due to Looseness/Weakness.....	2-42
2.734	Phase Behavior Due to Misalignment.....	2-42
2.735	Using Phase Analysis to Find the Operating Deflection Shape of a Machine and Its Support Frame.....	2-44
3.	OVERVIEW OF THE STRENGTHS AND WEAKNESSES OF TYPICAL VIBRATION INSTRUMENTS.....	3-1
3.0	Introduction.....	3-1
3.1	Instrument Comparisons.....	3-1
3.2	General Capabilities of Each Vibration Instrument Type.....	3-5
3.21	Overall Level Vibration Meters.....	3-5
3.22	Swept-Filter Analyzers.....	3-6
3.23	FFT Programmable Data Collectors.....	3-6
3.24	Real-Time Spectrum Analyzers.....	3-7
3.25	Instrument Quality Tape Recorders.....	3-8
4.	OVERVIEW OF VARIOUS VIBRATION TRANSDUCERS AND HOW TO PROPERLY SELECT THEM.....	4-1

SECTION	SUBJECT	PAGE
4.0	Introduction.....	4-1
4.1	Types of Vibration Transducers and Their Optimum Applications.....	4-1
4.11	Accelerometers.....	4-4
4.12	Velocity Pickups.....	4-9
4.13	Non-Contact Eddy Current Displacement Probes.....	4-14
4.14	Shaft Contact Displacement Probes.....	4-18
	4.141 Shaft Sticks.....	4-18
	4.142 Shaft Riders.....	4-20
4.2	Selection Criteria For Transducers.....	4-21
4.3	Mounting of Transducers.....	4-23
4.31	Transducer Mounting Applications.....	4-23
	Appendix - Specifications for Various Transducers from a Variety of Manufacturers.....	4-26
5.	ROLE OF SPIKE ENERGY, HFD AND SHOCK PULSE (SPM) AND SPECIFICATION OF THEIR ALARM LEVELS AT VARIOUS SPEEDS.....	5-1
A.	Spike Energy and Shock Pulse.....	5-1
6.	USE OF VIBRATION SIGNATURE ANALYSIS TO DIAGNOSE MACHINE PROBLEMS.....	6-1
6.0	Use of Vibration Signature Analysis.....	6-1
	Overview of 5-Page "Illustrated Vibration Diagnostic Chart".....	6-4
6.01	Mass Unbalance.....	6-12
6.011	Force Unbalance.....	6-15
6.012	Couple Unbalance.....	6-15
6.013	Dynamic Unbalance.....	6-16
6.014	Overhung Rotor Unbalance.....	6-17
	6.0141 Summary of Procedures for Balancing Overhung Rotors.....	6-17
6.015	Allowable Residual Unbalance and ISO Balance Quality Grade.....	6-21
6.02	Eccentric Rotors.....	6-27
6.03	Bent Shaft.....	6-30
6.04	Misalignment.....	6-32
6.041	Angular Misalignment.....	6-35
6.042	Parallel Misalignment.....	6-36
6.043	Misaligned Bearing Cocked on the Shaft.....	6-37
6.044	Coupling Problems.....	6-37
6.05	Natural Frequencies and Resonance.....	6-39
6.051	Natural Frequency.....	6-39
6.052	Resonance.....	6-42
6.06	Mechanical Looseness.....	6-43
6.061	Type A - Structural Frame/Base Looseness (1X RPM).....	6-43
6.062	Type B - Looseness Due to Rocking Motion or Cracked Structure/Bearing Pedestal (2X RPM).....	6-46
	Type C - Loose Bearing in Housing or Improper Fit Between Component Parts (Multiple Harmonics Due to Nonlinearity Often Induced By Impulse Events).....	6-48
6.07	Tracking of Rolling Element Bearing Failure Stages Using Vibration and High Frequency Enveloping and Demodulated Spectral Techniques.....	6-52
6.071	Optimum Vibration Parameter For Rolling Element Bearing Condition Evaluation (Acceleration, Velocity or Displacement).....	6-55
6.072	Types of Vibration Spectra Caused by Defective Rolling Bearings.....	6-57

<u>SECTION</u>	<u>SUBJECT</u>	<u>PAGE</u>
	6.073 Typical Spectra for Tracking Failure Stages Through Which Rolling Element Bearings Pass.....	6-70
6.08	Introduction to Gear Problem Detection.....	6-80
	6.081 Specification of Spectral Setup for Detecting Gear Wear.....	6-80
	6.082 Indications of Gear Tooth Wear.....	6-82
6.09	Introduction to Electrical Problem Detection.....	6-86
	6.091 Why Many Electrical Problems Occur at 2X Line Frequency.....	6-86
	6.092 Stator Problems.....	6-87
	6.093 Eccentric Rotor and Variable Air Gap.....	6-91
	6.094 Rotor Problems.....	6-93
	6.095 Important Closing Comments on Electrical Measurements.....	6-95
6.10	Belt Drive Problems.....	6-99
	6.101 Worn, Loose or Mismatched Belts.....	6-100
	6.102 Belt/Sheave Misalignment.....	6-102
	6.103 Eccentric Sheaves.....	6-102
	6.104 Belt Resonance.....	6-103
	6.105 Excessive Motor Vibration at Fan Speed Due to Motor Frame/Foundation Resonances.....	6-103
	6.106 Loose Pulley or Fan Hub.....	6-103
7.	PROVEN METHOD FOR SPECIFYING SPECTRAL BAND ALARM LEVELS AND FREQUENCIES USING TODAY'S PREDICTIVE MAINTENANCE SOFTWARE SYSTEMS.....	7-1
7.0	Abstract.....	7-1
7.1	Introduction to Specifying Spectral Alarm Bands & Frequency Ranges.....	7-2
	7.11 Two Types of Spectral Alarm Bands.....	7-3
	7.12 Which Vibration Parameter to Use in Spectral Alarm Bands - Displacement, Velocity or Acceleration?.....	7-4
	7.13 Review of Problems Detectable by Vibration Analysis.....	7-5
	7.14 Specification of Overall Vibration alarm Levels and Explanation of the Origin of Table II "Overall Condition Rating" Chart.....	7-13
	7.15 Specification of Spectral Alarm Levels and Frequency Bands Using Table III.....	7-16
	Case A - General Rolling Element Bearing Machine Without Rotating Vanes: (Motors, Spindles, Gearbox Lower Frequency Measurements, etc.).....	7-17
	Case B - General Sleeve Bearing Machine Without Rotating Vanes: (Sleeve Bearing Motors, Gearbox Lower Frequency Measurements, etc.).....	7-18
	Case C - Gearbox High Frequency Points with Known Number of Teeth.....	7-19
	Case D - Gearbox High Frequency Points with Unknown Number of Teeth.....	7-20
	Case E - Motor Electrical Rotor Bar Pass Frequency Point (Single Point Usually Taken on Outboard Motor Bearing).....	7-21
	Case F - Motor Electrical 12,000 CPM Measurement Point (Single Point Usually Taken on Inboard Motor Bearing).....	7-21
	Case G - Special Machine Types.....	7-22
	Type 1 - Centrifugal Machines with Known Number of Vanes (or Blades) and Rolling Element Bearings.....	7-22
	Type 2 - Centrifugal Machines with Unknown Number of Vanes (or Blades) and Rolling Element Bearings.....	7-22
	Type 3 - Centrifugal Machines with Known Number of Vanes (or Blades) and Sleeve Bearings.....	7-23
	Type 4 - Centrifugal Machines with Unknown Number of Vanes (or Blades) and Sleeve Bearings.....	7-23
	7.151 Examples - Specification of Spectral Alarm Bands for Sample Machines.....	7-23

SECTION	SUBJECT	PAGE
7.16	Periodic Reevaluation of Spectral alarm Band Setups on Each Family of Machines.....	7-29
7.161	Procedure for Evaluating the Effectiveness of Specified Overall Alarm Levels and Spectral Bands.....	7-31
7.162	Example - "Statistical Analysis of Overall Vibration Velocity in 4 Client Power Plants Using the Procedure Recommended Above".....	7-32
7.17	Conclusions.....	7-34
8.	COMMON PITFALLS IN EVERYDAY VIBRATION MEASUREMENTS.....	8-1
8.0	Introduction.....	8-1
8.1	General Consideration for Obtaining Consistent Quality Data.....	8-1
8.11	Choosing Measurement Locations.....	8-1
8.12	Machine and Point Identification.....	8-3
8.13	Measurement Parameters.....	8-7
8.14	Instrument Selection, Setup, and Condition.....	8-12
8.15	Measurement Techniques.....	8-14
8.16	Transducer Mounting and Probes.....	8-15
8.2	Effect of Transducer Mounting on Vibration Measurements.....	8-19
9.	SETUP AND IMPLEMENTATION OF PREDICTIVE MAINTENANCE AND CONDITION MONITORING PROGRAMS.....	9-1
•	"Flow Chart of Recommended Predictive Maintenance Programs".....	9-7
•	"Sample PMP Machine Plant Layout".....	9-8
•	"Sample Machinery Drawing and Configuration Chart".....	9-9
•	"Criteria for Overall Condition Rating (Peak Overall Velocity, in/sec)".....	9-10
•	"Sample Overall Condition Rating Report".....	9-12
•	"Sample Overall Condition Rating Bar Graph".....	9-14
•	"Sample Rank-Ordered Results and Recommendations Report".....	9-16
•	"Sample PMP Machine Repair Log".....	9-17
•	"Latest Overall Measurement Report".....	9-19
•	"Overall Alarm Exception Report".....	9-21
•	"Spectral Band Alarm Report".....	9-23
•	"Spectral Narrowband Envelope Alarm Report".....	9-34
•	"Current Inspection Code Report".....	9-44
•	"Trends and Waterfall Plots".....	9-45
•	"Effect of Vibration Acceptance Testing".....	9-49
10.	REAL-WORLD CASE HISTORIES.....	10-1

(See Individual Table of Contents At Beginning of This Section)

RECOMMENDED PERIODICALS FOR THOSE INTERESTED IN PREDICTIVE MAINTENANCE

1. Sound and Vibration Magazine

P.O. Box 40416
Bay Village, OH 44140
Mr. Jack Mowry, Editor and Publisher
Phone: 216-835-0101
Fax : 216-835-9303

Terms: Normally free for bona fide qualified personnel concentrating in the Sound and Vibration Analysis/Plant Engineering Technologies. Non-qualified personnel - \$25/per year within the U.S.

Comments: This is a monthly publication that normally will include approximately 4-6 issues per year devoted to Predictive Maintenance. Their Predictive Maintenance articles are usually practical and in good depth; normally contain real "meat" for the PPM vibration analyst. Sound and Vibration has been published for over 25 years.

2. Vibrations Magazine

The Vibration Institute
6262 South Kingery Hwy, Suite 212
Willowbrook, IL 60514
Institute Director - Dr. Ronald Eshleman
Phone: 630-654-2254
Fax : 630-654-2271

Terms: Vibrations Magazine is sent to Vibration Institute members as part of their annual fee, (approx. \$45 per year). It is available for subscription to non-members at \$55/per year; \$60/foreign.

This is a quarterly publication of the Vibration Institute. Always contains very practical and useful Predictive Maintenance Articles and Case Histories. Well worth the small investment.

Comments: Yearly Vibration Institute fee includes reduced proceedings for that year if desired for the National Conference normally held in June. They normally meet once per year at a fee of about \$675/per person, (\$600/person for Institute members) including conference proceedings notes and mini-seminar papers. All of the papers presented, as well as mini-courses, at the meeting are filled with "meat" for the Predictive Maintenance Vibration Analyst. Vibrations Magazine was first published in 1985 although the Institute has been in existence since approximately 1972, with their first annual meeting in 1977. The Vibration Institute has several chapters located around the United States which normally meet on a quarterly basis. The Carolinas' Vibration Institute Chapter normally meets in Greenville, SC; Charleston, SC; Columbia, SC; Charlotte, NC; Raleigh, NC; and in the Winston Salem, NC areas. For Institute membership information, please contact: Dr. Ron Eshleman at 630-654-2254. When doing so, be sure to ask what regional chapter is located to your area. Membership fees for the "Annual Meeting Proceedings" are \$30/per year (normal cost is approx. \$60/per year for proceedings if annual meeting is not attended). Please tell Ron that we recommended you joining the Vibration Institute when you call or write to him.

R-0697-1

3. P/PM Technology Magazine

P.O. Box 1706

Minden, NV 89423-1706 (Pacific Coast Time)

Phone: 702-267-3970; 800-848-8324

Fax : 702-267-3941

Publisher- Mr. Ronald James; Assistant Publisher: Susan Estes

Terms: \$42/per year for qualified USA subscribers, (individuals and establishments involved with industrial plant and facilities maintenance; subscribers must be associated in engineering, maintenance, purchasing or management capacity). \$60/year for unqualified subscribers.

Comments: This is a bi-monthly magazine with articles about all facets of PPM Technologies, including Vibration Analysis, Oil Analysis, Infrared Thermography, Ultrasonics, Steam Trap Monitoring, Motor Current Signature Analysis, etc. These are normally good practical articles. Also includes some cost savings information, although does not necessarily include how these cost savings were truly determined. *P/PM Technology* also hosts at least one major conference per year in various parts of the United States. Intensive training courses in a variety of condition monitoring technologies will also be offered in vibration analysis, root cause failure analysis, oil analysis, thermographic analysis, ultrasonic analysis, etc..)

4. Maintenance Technology Magazine

1209 Dundee Ave., Suite 8

Elgin, IL 60120

Phone: 800-554-7470

Fax : 804-304-8603

Publisher: Arthur L. Rice

Terms: \$95/per year for non-qualified people This is a monthly magazine that usually has at least one article relating to Predictive Maintenance using vibration analysis within each issue. In addition to vibration, it likewise always offers other articles covering the many other technologies now within Predictive Maintenance.

5. Reliability Magazine

PO Box 856

Monteagle, TN 37356

Phone: 423-592-4848

Fax : 423-592-4849

Editor: Mr. Joseph L. Petersen

Terms: \$49 per year in USA; \$73 per year outside USA.

Comments: This bi-monthly magazine covers a wide variety of Condition Monitoring Technologies including Vibration Analysis, Training, Alignment, Infrared Thermography, Balancing, Lubrication Testing, CMMS and a unique category they entitle "Management Focus".

NOTE: In addition to these periodicals, many of the major predictive maintenance hardware and software vendors put out periodic newsletters. Some of these in fact do include some "real meat" in addition to their sales propaganda. We would recommend that you contact, particularly the vendor supplying your predictive maintenance system for their newsletter. Their newsletter will likewise advise you of updates in their current products.

ANALYSIS I SEMINAR AGENDAS

INTRODUCTION TO VIBRATION ANALYSIS AND PREDICTIVE MAINTENANCE

(Recommended Full-time PMP Vibration Experience = 0 to 6 Months)

1. Seminar Overview
2. What is Vibration and How Can it be Used to Evaluate Machinery Condition?
 - What is Frequency and How Does it Relate to a Time Waveform?
 - What is Vibration Displacement, Velocity, and Acceleration?
 - How to Convert From One Vibration Parameter to Another (in/sec to g, etc.)
 - What is Phase?
 - What is a Vibration Spectrum (Also Called an "FFT" or "Signature")?
 - Effect on Frequency Accuracy of No. of FFT Lines
 - Difference Between RMS, Peak, and Peak-To-Peak Amplitude and its Significance
 - When to Use Displacement, Velocity, or Acceleration
 - How Much is Too Much Vibration?
3. Overview of the Strengths and Weaknesses of Typical Vibration Instruments
4. Overview of Various Vibration Transducers and Their Optimum Applications
5. Role of Spike Energy, High Frequency Detection (HFD), and Shock Pulse, along with Proper Specification of Their Alarm Levels at Various Speeds
6. Introduction to Vibration Signature Analysis and How it is Used to Evaluate Machine Operating Condition
Overview of 5-Page "Illustrated Vibration Diagnostic Chart"
 - Mass Unbalance
 - Eccentric Rotors
 - Bent Shaft
 - Misalignment
 - Mechanical Looseness, Improper Component Fit, and Soft Foot
 - Tracking of Rolling Element Bearing Health Using Vibration Signature Analysis
 - Belt Drive Problems
 - Electrical Problems in Motors
7. Proven Method for Specifying Spectral Band Alarm Levels and Frequencies Using Today's Predictive Maintenance Software Systems
8. Common Pitfalls in Everyday Vibration Measurements and Their Effect on Properly Detecting and Diagnosing Machine Problems
 - Proper and Improper Measurement Locations
 - Effect of Transducer Mounting on Resulting Vibration Spectra
 - Incorrect Transducer Choice
 - Incorrect Measurement Parameter (mils, in/sec, g, gSE, etc.)
 - Incorrect Frequency Span
 - Poor Measurement Techniques
9. Setup and Implementation of Effective Predictive Maintenance and Condition Monitoring Programs
10. Actual Case Histories of Vibration Diagnostics on Various Machine Types

ANALYSIS II SEMINAR AGENDAS

CONCENTRATED VIBRATION SIGNATURE ANALYSIS AND RELATED CONDITION MONITORING TECHNIQUES

(Recommended Full-time PMP Vibration Experience = 6 to 12 Months)

1. Seminar Overview
2. Brief Review of Analysis I Seminar Topics
3. Proper Use of Vibration Severity Charts
4. How to Refine Overall Alarm and Spectral Band Alarm Levels after Several surveys have been taken
5. Phase Analysis Techniques
6. Detailed Vibration Signature Analysis for 40 Machine Problems Using Technical Associates' 5-Page "**Illustrated Vibration Diagnostics Chart**":
 - Mass Unbalance
 - Eccentric Rotors
 - Bent Shaft
 - Misalignment and Coupling Problems
 - Machinery Failures Due to Resonant Vibration
 - Mechanical Looseness - 3 Distinctly Different Types
 - Rotor Rub
 - Sleeve Bearing Problems
 - Rolling Element Bearing Problems
 - Flow-Induced Vibration (Cavitation, Starvation, etc..)
 - Gear Problems
 - Electrical Problems in Motors
 - Belt-Drive Problems
 - Beat Vibration Problems
 - Soft Foot and Distorted Frame Problems
7. Presentation of a Proven Analysis Procedure Using the "**5-Page Illustrated Vibration Diagnostic Chart**" (Including Examples)
8. What is "High Frequency Enveloping and Demodulation Spectral Analysis" (Used to Detect Problems with Rolling Element Bearings, Gears, Lubrication, etc.)?
9. How to Select the Optimum Condition Monitoring Hardware and Software Using Proven Weighted Matrix Methods Based on Cost, Hardware Capabilities, Software Capabilities and Vendor History (Generic Procedure Independent of PMP Vendor)
10. Presentation of **Real-World Case Histories**

ANALYSIS III SEMINAR AGENDAS

INTRODUCTION TO SPECIAL VIBRATION ANALYSIS TECHNIQUES AND NARROWBAND SPECTRAL ENVELOPE ALARMS

(Recommended Full-time PMP Vibration Experience = 12 to 24 Months)

1. Seminar Overview
2. Review of Analysis II Seminar Topics
3. Introduction to Frequency and Amplitude Interpolation and Why They are so Important to Accurate Vibration Signature Analysis
4. How Can One Diagnose a Bearing, Gear or Electrical Problem When He Does Not Know Rolling Element Bearing Model Numbers, Gear Tooth Count, or Number of Rotor Bars?
5. How to Properly Create and Specify Optimum Narrowband Envelope Spectral Alarms for Various "Families" of Machines Having Different Operating Speeds
6. How Much is Too Much at Specific Vibration Frequencies for Various Machine Types?
 - At Operating Speed (1x RPM)
 - At 2x RPM
 - At 3x RPM
 - At 2x Line Frequency in an Induction Motor
 - At 1x and 2x Rotor Bar Pass Frequency in an Induction Motor
 - At Blade Pass Frequency and Harmonics
 - At Gear Mesh Frequency and Harmonics
 - At Belt Frequencies and Harmonics
 - At Rolling Element Bearing Frequencies and Harmonics
 - At SCR Firing Frequencies in DC Motors
 - At Other Electrical Line Frequencies in DC Motors
 - At Coil Passing Frequency in Synchronous Motors
7. Acceptance Tests and Criteria for New and/or Overhauled Machinery
8. How to Accurately Track Rolling Element Bearing Condition Using Both Vibration Signature Analysis and High Frequency Enveloping and Demodulated Spectral Techniques
9. Introduction to Gear Problem Diagnostics
10. Natural Frequency Testing
 - Impulse Natural Frequency Tests
 - Coastdown and Runup Tests
 - Bode and Nyquist Plotting (Phase and Amplitude vs. RPM)
 - Swept-sine Variable Frequency Shakers
11. Introduction to Special Diagnostic Techniques
 - Low Frequency Vibration Measurement Considerations (Excerpts from "Advanced 1" Seminar Text)
 - Operating Deflected Shape Analysis (ODS) by Manual as well as by Automated Software-driven Methods
 - Time Waveform Analysis and how it Relates to FFT Analysis
 - Synchronous Time Averaging (Excerpts from "Advanced I" Seminar Text)
 - Introduction to Spectral Analysis of Motor Current

12. Presentation of Effective PMP Reporting Techniques to Keep Program Successes Visible and to Ensure Proper Corrective Actions are Taken
13. Considerations for Vibration Analysis of Specific Machine Types:
 - Horizontal Pumps
 - Vertical Pumps
 - Centrifugal Air Compressors
 - Centrifugal Fans
14. Presentation of **Real-World Case Histories**

ADVANCED IV SEMINAR AGENDAS

ADVANCED VIBRATION DIAGNOSTIC TECHNIQUES AND HOW TO PROPERLY EVALUATE LOW, HIGH, AND VARIABLE SPEED MACHINES

(Recommended Full-Time PMP Vibration Experience = 24 to 48 Months)

1. Seminar Overview
2. Review of Analysis III Seminar Topics
3. More on the High Frequency Enveloping and Demodulation Techniques
 - What it Means
 - Setting Up the Parameters (Filters, Frequency Spans, etc.)
 - How to Interpret the Results
 - Examples of High Frequency Enveloping Spectra ,
4. Special Analysis Techniques
 - Time Waveform Analysis - When and Where it Detects Special Problems FFT Signature Analysis Misses
 - Synchronous Time Averaging - How it is Used to Pinpoint Problems Other Techniques Cannot
 - Order Tracking - Why it is Especially Important to Properly Evaluate Variable-Speed Machines
 - Motor Current Analysis and Presentation of a "Motor Current Troubleshooting and Severity Chart" Developed by Technical Associates
 - Advanced Natural Frequency Testing and Confirmation of What Component is Resonant
 - Introduction to Experimental Modal Analysis Using Multi-channel Analyzers
 - Use of Real-Time Analyzers (Single and Multi-Channel)
5. Low-Speed Machine Vibration Analysis Techniques and Instrumentation Requirements (Predominantly Below 100 RPM)
6. High-Speed Machine Vibration Analysis Techniques, Mounting and Instrumentation Requirements (Generating Frequencies Exceeding 10,000 Hz or 600,000 CPM)
7. How to Analyze Variable Speed Machines Both on a One-Time Diagnostic Basis as well as in a Condition Monitoring Program
8. Gaining a Better Understanding of What Today's Vibration Analyzers are Doing in Practical Terms
 - Significance of Dynamic Range and Number of A/D Bits
 - Significance of Measurement Bandwidth (i.e., "How Far Apart do Two Peaks Have to be for an FFT Analyzer to Display Two Separate Frequencies"?)
 - What are the Real Advantages and Disadvantages of Using Various FFT Windows (i.e., Hanning, Flatop, Uniform, etc.)?
 - Overlap Processing - What is it, Along with its Advantages and Disadvantages?
 - Interpolated Frequency Enhancement - What is it and Why is it so Significant to Accurate Machinery Problem Diagnostics?
9. Presentation of **Real-World Case Histories**



TECHNICAL ASSOCIATES PRE-SEMINAR QUESTIONNAIRE

YOUR NAME: _____ SEMINAR DATES: _____
TITLE: _____ SEMINAR LOCATION: _____
COMPANY: _____ SEMINAR LEVEL (CIRCLE) BAL ENTRY I II III ADV MODAL
LOCATION: _____ VIBRATION EXPERIENCE: YRS. _____ MOS. _____
PHONE: _____ EXT: _____ SPECTRAL ANAL. EXPER: YRS. _____ MOS. _____
EMAIL: _____ FAX: _____

PLEASE COMPLETE THE FOLLOWING QUESTIONNAIRE, THIS INFORMATION IS FOR USE IN THIS CLASS AND TO HELP US PLAN FUTURE SEMINARS: (SEE SAMPLE PRE-SEMINAR SURVEY ENCLOSED)

A. TYPE OF INDUSTRY AND PRODUCT MANUFACTURED/SERVICES RENDERED BY YOUR COMPANY:

AUTOMOTIVE _____	CHEMICAL _____	CONSUMER PRODS. _____
CONSULTANT _____	FOOD/BEVERAGES _____	FABRICA. METAL _____
FIBERGLASS _____	HVAC _____	INSTRUM./S'WARE _____
MINING _____	PETROLEUM _____	PHARMACEUTICAL _____
PULP & PAPER _____	REPAIR _____	SYNTHETIC FIBER _____
TEXTILES _____	TOBACCO _____	UTILITIES _____
VIB. SERVICES _____	TRAINING SERVICES _____	OTHER (SPECIFY) _____

B. COMPANY INFORMATION:

PRIMARY PRODUCT(S) AT YOUR FACILITY? _____
NUMBER OF EMPLOYEES AT YOUR FACILITY? _____
DO YOU HAVE AN ONGOING VIBRATION/PdM PROGRAM? YES _____ NO _____
IF YES, HOW LONG HAS PROGRAM BEEN IN PLACE? _____ YRS.
PERSON RESPONSIBLE FOR PROGRAM? NAME: _____
TITLE: _____

C. WHAT TYPE OF VIBRATION INSTRUMENTS ARE USED AT YOUR PLANT? (ENTER BOTH MANUFACTURER MODEL NO. AND QUANTITY: i.e., CSI 2130 (3))

• OVERALL LEVEL METER _____ (i.e., SKF Pen Plus)	• SWEPT-FILTER ANALYZER _____ (i.e., SCHENCK VIBRAPORT 30)
• DATA COLLECTOR _____ (i.e., Entek DataPAC 1500)	• 2 CHANNEL RTA _____ (i.e., HP 35670)
• MULTI-CHANNEL DATA _____ COLLECTOR (i.e., CSI 2130)	• MULTI-CHANNEL RTA _____ (i.e., SA 395)

D. WHAT IS THE BRAND NAME AND MODEL # OF THE PRIMARY DATA COLLECTOR USED AT YOUR PLANT? (INDICATE MODEL NO. IN BLANK: i.e., SKF CMVA-60)

BENTLY _____	COMMTTEST _____	CSI _____	DI _____	DLI _____
ENTEK _____	PRUFTECHNIK _____	SKF _____	OTHER _____	

E. WHAT PMP SOFTWARE AND VERSION # DO YOU USE? (INDICATE ALL SOFTWARE THAT YOU NOW USE)

ALERT _____	ASCENT _____	IQ 2000 _____	MACHINE ANALYST _____
MASTERTREND _____	ODYSSEY _____	OMNITREND _____	PRISM ⁴ _____
RBMWARE _____	TRENDMASTER _____		OTHER _____

F. EXPERT DIAGNOSTIC SOFTWARE USED AT YOUR PLANT: (PLACE CHECK MARK)

EXPERTALERT _____ MDU _____ MOTORCHECK _____
MOTORMONITOR _____ NSPECTR II _____ PRISM4 PRO _____ ROMEX _____
OTHER (PLEASE IDENTIFY) _____ NONE _____

G. EXPERIMENTAL MODAL ANALYSIS SOFTWARE USED AT YOUR PLANT (CHECK ALL THOSE WHICH YOU NOW USE).

LMS _____ ME'scope _____ SDRC _____ STAR _____ ZONIC _____ OTHER (SPECIFY) _____

H. HAVE YOU PERFORMED FINITE ELEMENT MODELING? IF SO, WITH WHAT SOFTWARE?

YES _____ NO _____ SOFTWARE USED? _____

I. WHAT OTHER VIBRATION/PREDICTIVE MAINTENANCE SEMINARS HAVE YOU ATTENDED? SEMINAR NAME SEMINAR COMPANY APPROX. DATE ATTENDED

J. IS THERE ANY SPECIAL MACHINE TYPE IN YOUR FACILITY ON WHICH YOU NEED HELP? AND/OR WOULD LIKE ADDITIONAL INFORMATION?

MACHINE NAME BEARING TYPE RPM HP DRIVE TYPE

K. PLEASE LIST ANY SPECIAL QUESTIONS OR TOPICS THAT YOU WOULD LIKE COVERED DURING THIS SEMINAR?

L. HOW DID YOU FIND OUT ABOUT THIS SEMINAR? (PLEASE CHECK PRIMARY CONTACT)

BROCHURE _____ DIRECT SALES _____ PHONE CALL _____
COLLEAGUE _____ FORMER ATTENDER _____ TA WEB SITE _____
MAGAZINE AD (If So, Which Magazine)? _____ OTHER (Please Indicate Source) _____

M. WHAT PROMPTED YOU TO ATTEND THIS PARTICULAR SEMINAR? (PLEASE CHECK ALL THAT APPLY.)

ADVANCEMENT _____ SEMINAR LOCATION _____ SPEAKER _____
SUBJECT MATTER _____ WAS RECOMMENDED _____ OTHER _____

PLEASE SUBMIT THIS COMPLETED FORM AND MAIL/FAX TO TECHNICAL ASSOCIATES BEFORE ATTENDING SEMINAR. THANK YOU. WE APPRECIATE YOUR TAKING THE TIME TO HELP US CUSTOMIZE AND ENHANCE THIS SEMINAR TO ENSURE WE MEET PARTICULAR NEEDS.

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

SAMPLE

SAMPLE

TECHNICAL ASSOCIATES PRE-SEMINAR QUESTIONNAIRE

YOUR NAME: John W. Doe SEMINAR DATES: June 8-11, 2004
 TITLE: Vibration Analyst I SEMINAR LOCATION: Myrtle Beach, SC
 COMPANY: XYZ Chemicals, Inc. SEMINAR LEVEL (CIRCLE) BAL ENTRY I III ADV MODAL
 LOCATION: Americana, TX VIBRATION EXPERIENCE: YRS. 3 MOS. 0
 PHONE: 512-222-1234 EXT: 765 SPECTRAL ANAL. EXPER: YRS. 1 MOS. 6
 EMAIL: JWDoe@XYZchemicals.com FAX: 512-212-1111

PLEASE COMPLETE THE FOLLOWING QUESTIONNAIRE, THIS INFORMATION IS FOR USE IN THIS CLASS AND TO HELP US PLAN FUTURE SEMINARS: (SEE SAMPLE PRE-SEMINAR SURVEY ENCLOSED)

A. TYPE OF INDUSTRY AND PRODUCT MANUFACTURED/SERVICES RENDERED BY YOUR COMPANY:

AUTOMOTIVE _____	CHEMICAL _____ ☒	CONSUMER PRODS. _____
CONSULTANT _____	FOOD/BEVERAGES _____	FABRICA. METAL _____
FIBERGLASS _____	HVAC _____	INSTRUM./SWARE _____
MINING _____	PETROLEUM _____	PHARMACEUTICAL _____
PULP & PAPER _____	REPAIR _____	SYNTHETIC FIBER _____
TEXTILES _____	TOBACCO _____	UTILITIES _____
VIB. SERVICES _____	TRAINING SERVICES _____	OTHER (SPECIFY) _____

B. COMPANY INFORMATION:

PRIMARY PRODUCT(S) AT YOUR FACILITY? Industrial Chemicals
 NUMBER OF EMPLOYEES AT YOUR FACILITY? 1600
 DO YOU HAVE AN ONGOING VIBRATION/PdM PROGRAM? YES ☒ NO _____
 IF YES, HOW LONG HAS PROGRAM BEEN IN PLACE? 7 YRS.
 PERSON RESPONSIBLE FOR PROGRAM? NAME: Tom Smith
 TITLE: Reliability Engineer

C. WHAT TYPE OF VIBRATION INSTRUMENTS ARE USED AT YOUR PLANT?

(ENTER BOTH MANUFACTURER MODEL NO. AND QUANTITY: i.e., CSI 2130 (3))

• OVERALL LEVEL METER <u>SKF PenPlus(3)</u> (i.e., SKF Pen Plus)	• SWEPT-FILTER ANALYZER <u>IRP 880(1)</u> (i.e., SCHENCK VIBRAPORT 30)
• DATA COLLECTOR <u>SKF CMVA 60(2)</u> (i.e., Entek DataPAC 1500)	• 2 CHANNEL RTA _____ (i.e., HP 35670)
• MULTI-CHANNEL DATA <u>CSI 2130(4)</u> COLLECTOR (i.e., CSI 2130)	• MULTI-CHANNEL RTA <u>Iotech ZonicBook</u> (i.e., SA 395)

D. WHAT IS THE BRAND NAME AND MODEL # OF THE PRIMARY DATA COLLECTOR USED AT YOUR PLANT? (INDICATE MODEL NO. IN BLANK: i.e., SKF CMVA-60)

BENTLY _____	COMMTEST _____	CSI <u>2130</u>	DI _____	DLI _____
ENTEK IRD _____	PRUFTECHNIK _____	SKF _____	OTHER _____	

E. WHAT PMP SOFTWARE AND VERSION # DO YOU USE? (INDICATE ALL SOFTWARE THAT YOU NOW USE)

ALERT _____	ASCENT _____	IQ 2000 _____	MACHINE ANALYST <u>V 2.2</u>
MASTERTREND _____	ODYSSEY _____	OMNITREND _____	PRISM ⁴ _____
RBWARE <u>v. 4.81</u>	TRENDMASTER _____	___ OTHER _____	

F. EXPERT DIAGNOSTIC SOFTWARE USED AT YOUR PLANT: (PLACE CHECK MARK)

EXPERTALERT _____ MDU _____ MOTORCHECK _____
 MOTORMONITOR _____ NSPECTR II ☒ PRISM4 PRO _____ ROMEX _____
 OTHER (PLEASE IDENTIFY) _____ NONE

G. EXPERIMENTAL MODAL ANALYSIS SOFTWARE USED AT YOUR PLANT (CHECK ALL THOSE WHICH YOU NOW USE).

LMS _____ ME'scope ☒ SDRC _____ STAR _____ ZONIC _____ OTHER (SPECIFY) _____

H. HAVE YOU PERFORMED FINITE ELEMENT MODELING? IF SO, WITH WHAT SOFTWARE?

YES _____ NO ☒ SOFTWARE USED? _____

WHAT OTHER VIBRATION/PREDICTIVE MAINTENANCE SEMINARS HAVE YOU ATTENDED?

SEMINAR NAME	SEMINAR COMPANY	APPROX. DATE ATTENDED
Entry Level	Technical Associates	Feb., 2001
Dynamic Balancing	Technical Associates	June, 2001
Analysis I	Technical Associates	June, 2003

J. IS THERE ANY SPECIAL MACHINE TYPE IN YOUR FACILITY ON WHICH YOU NEED HELP? AND/OR WOULD LIKE ADDITIONAL INFORMATION?

MACHINE NAME	BEARING TYPE	RPM	HP	DRIVE TYPE
Hoffman Blower (7-stage)	R.E. Bearing	3580	150 HP	Direct Coupled
Buffalo Forge Scrubber Fan	R.E. Bearing	1180	100 HP	Belt Drive

K. PLEASE LIST ANY SPECIAL QUESTIONS OR TOPICS THAT YOU WOULD LIKE COVERED DURING THIS SEMINAR?

1) How can you reliably diagnose problems with Rolling Element Bearings?
 2) Is there now any technique available to detect problems on an intermediate gear inside a 4-stage Gearbox?
 3) What is PeakVue and how should I set it up for field measurements?

L. HOW DID YOU FIND OUT ABOUT THIS SEMINAR? (PLEASE CHECK PRIMARY CONTACT)

BROCHURE ☒ DIRECT SALES _____ PHONE CALL _____
 COLLEAGUE _____ FORMER ATTENDER ☒ TA WEB SITE ☒
 MAGAZINE AD (If So, Which Magazine)? _____ OTHER (Please Indicate Source) _____

M. WHAT PROMPTED YOU TO ATTEND THIS PARTICULAR SEMINAR? (PLEASE CHECK ALL THAT APPLY.)

ADVANCEMENT ☒ SEMINAR LOCATION _____ SPEAKER _____
 SUBJECT MATTER _____ WAS RECOMMENDED ☒ OTHER _____

PLEASE SUBMIT THIS COMPLETED FORM AND MAIL/FAX TO TECHNICAL ASSOCIATES BEFORE ATTENDING SEMINAR, THANK YOU. WE APPRECIATE YOUR TAKING THE TIME TO HELP US CUSTOMIZE AND ENHANCE THIS SEMINAR TO ENSURE WE MEET PARTICULAR NEEDS.

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.



TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

PAGE 1

SINCE 1961

347 N. Caswell Road

Charlotte, NC 28204

Ph: (704) 333-9011

Fax: (704) 333-1728

www.technicalassociates.net

PUBLICATION ORDER FORM

QTY. PRICE TOTAL

ILLUSTRATED VIBRATION DIAGNOSTIC WALL CHART

DWC-4 (LAMINATED 38" X 52")

\$ 99.00

DWCSP-4 (LAMINATED 38" X 52") Spanish Version **NEW**

\$ 99.00

USE OF VIBRATION ANALYSIS TO DETECT A VARIETY OF MACHINE PROBLEMS

USING TECHNICAL ASSOCIATES' "ILLUSTRATED VIBRATION DIAGNOSTIC WALL CHART"

DWD-4 (183 PAGE BOOKLET)

\$ 199.00

BOTH THE "ILLUSTRATED VIBRATION DIAGNOSTIC WALL CHART" AND COMPLETE BOOKLET

DWE-4

\$ 259.00

POCKET DIAGNOSTIC WALL CHART **NEW**

(POCKET SIZED VERSION OF WORLD RENOWN ILLUSTRATED VIBRATION DIAGNOSTIC WALL CHART)

PDWC-4

\$49.00

INTERACTIVE VIBRATION DIAGNOSTIC CHART-ELECTRONIC VERSION OF THE "ILLUSTRATED VIBRATION DIAGNOSTIC WALL CHART"

IVDC-1 (CD ROM)

\$ 169.00

PROVEN METHODS FOR SPECIFYING BOTH 6 SPECTRAL BANDS AS WELL AS NARROWBAND

ALARM ENVELOPES USING TODAY'S PREDICTIVE MAINTENANCE SOFTWARE SYSTEMS - 4th EDITION

SAB-4 (57 PAGE BOOKLET ONLY)

\$ 79.00

DDT-1 DATABASE DEVELOPMENT TOOL SOFTWARE WITH "PROVEN METHODS" 4th EDITION (ON CD-ROM) **NEW**

\$ 299.00

VAT-8 VIBRATION DIAGNOSTICS TRAINING VIDEOTAPE SERIES (SET OF 7)

\$ 2,495.00

VWB-8 VIDEO WORKBOOKS (SET OF 7)

\$ 210.00

ADVANCED VIBRATION ANALYSIS - AC INDUCTION MOTORS TRAINING VIDEOTAPE SERIES

AVT-3 (SET OF 2)

\$ 995.00

ATB-1 VIDEO WORKBOOKS

\$ 39.00

BandAID SOFTWARE IS THE QUICKEST WAY TO DISPLAY INFORMATION REQUIRED FOR EACH MACHINE

POINT WHEN BUILDING A DATABASE

BFW-3 (50 PAGE BOOKLET AND WINDOWS SOFTWARE)

\$ 460.00

VIBRATION ACCEPTANCE STANDARDS USING BandAID

VAS-1 (60 PAGE BOOKLET)

\$ 99.00

HOW TO AND HOW NOT TO CREATE MEANINGFUL NARROWBAND SPECTRAL ALARM ENVELOPES

SAE-1 (53 PAGE BOOKLET)

\$ 79.00

TRACKING OF ROLLING ELEMENT BEARING FAILURE STAGES USING VIBRATION SIGNATURE ANALYSIS

REB-1 (90 PAGE BOOKLET)

\$ 79.00

REQUIRED VIBRATION ANALYSIS TECHNIQUES AND INSTRUMENTATION ON LOW-SPEED MACHINES

(PARTICULARLY 10 TO 300 RPM MACHINERY)

LSM-1 (95 PAGE BOOKLET)

\$ 79.00

REQUIRED VIBRATION ANALYSIS TECHNIQUES AND INSTRUMENTATION ON HIGH-SPEED MACHINES

(AT FREQUENCIES > 10,000 Hz OR 600,000 CPM)

HSM-1 (33 PAGE BOOKLET)

\$ 59.00

HOW TO ANALYZE VARIABLE-SPEED MACHINES AND INCLUDE THEM IN A PREDICTIVE

MAINTENANCE PROGRAM

VSM-1 (19 PAGE BOOKLET)

\$ 59.00

HIGH FREQUENCY ENVELOPING AND DEMODULATION TECHNIQUES - 3 YEAR STUDY

HFE-1 (61 PAGE BOOKLET)

\$ 59.00

HOW TIME WAVEFORM ANALYSIS AND TRUE SYNCHRONOUS TIME AVERAGING CAN BE USED TO

EFFECTIVELY EVALUATE MACHINES WHICH HAVE PROBLEMS NOT NECESSARILY DETECTABLE BY FFT

ANALYSIS ALONE

TWA-1 (29 PAGE BOOKLET)

\$ 59.00

DIAGNOSTIC TESTS AND INSTRUMENTATION REQUIRED FOR NATURAL FREQUENCY PROBLEM

DETECTION AND CORRECTION

NFT-1 (60 PAGE BOOKLET)

\$ 79.00

COMPARISON OF MOTOR CURRENT ANALYSIS AND VIBRATION ANALYSIS IN DETECTING ROTOR BAR AND

AIR GAP ECCENTRICITY PROBLEMS IN INDUCTION MOTORS

MCA-1 (31 PAGE BOOKLET)

\$ 49.00

DC MOTOR & CONTROL PROBLEM DETECTION USING VIBRATION ANALYSIS

DCM-1 (28 PAGE BOOKLET)

\$ 59.00

HOW TO DIAGNOSE PROBLEMS WITHIN 2-STAGE ROTARY SCREW AIR COMPRESSORS USING VIBRATION AND

HIGH FREQUENCY ENVELOPING ANALYSIS—2ND Edition **NEW**

RSC-2 (135 PAGE EDITION) [90 Special Offer-Aug 1-Oct 31, 2005 cost \$79.00]

\$129.00

TOTAL OF PAGE 1

PLEASE COMPLETE PAGE 2 OF ORDER FORM



TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

PAGE 2

SINCE 1961

347 N. Caswell Road

Charlotte, NC 28204

Ph: (704) 333-9011

Fax: (704) 333-1728

www.technicalassociates.net

PUBLICATION ORDER FORM

QTY. PRICE TOTAL

DESCRIPTION OF PEAKVUE & ILLUSTRATION WIDE ARRAY OF APPLICATIONS IN FAULT DETECTION &
PROBLEM SEVERITY ASSESSMENT

PKV-1 (95 PAGE BOOKLET)

\$ 99.00

HIGH-SPEED CENTRIFUGAL AIR COMPRESSOR DIAGNOSTICS

CAC-1 (14 PAGE BOOKLET)

\$ 39.00

ENTRY LEVEL VIBRATION ANALYSIS AND PREDICTIVE MAINTENANCE TECHNIQUES

VSA-0 (200 PAGE SEMINAR BOOK)

\$ 250.00

VIBRATION SIGNATURE ANALYSIS I

VSA-1 (410 PAGE SEMINAR BOOK)

\$ 300.00

VIBRATION SIGNATURE ANALYSIS II

VSA-2 (490 PAGE SEMINAR BOOK)

\$ 325.00

VIBRATION SIGNATURE ANALYSIS III

VSA-3 (510 PAGE SEMINAR BOOK)

\$ 350.00

ADVANCED VIBRATION DIAGNOSTIC AND REDUCTION TECHNIQUES

ADV-1 (1150 PAGE SEMINAR BOOK)

\$ 500.00

DYNAMIC FIELD BALANCING

DFB-1 (100 PAGE SEMINAR BOOK)

\$ 200.00

NOTE: THE ABOVE PRICES INCLUDE UPS SHIPPING AND HANDLING WITHIN THE USA

FOR DELIVERY IN N.C., ADD 7% (some counties 7.5%) SALES TAX OR ADD SHIPPING CHARGES OUTSIDE USA

TOTAL FROM PAGE 1

TOTAL FROM PAGE 2

TOTAL ORDER (US DOLLARS)

BILLING INFORMATION

DATE

INVOICE NUMBER

P.O. NUMBER:

CREDIT CARD NUMBER

☐ MASTERCARD ☐ VISA ☐ AMEX ☐ DISCOVER

EXP. DATE:

V-CODE: (Last set of digits on back of card on signature line)

CARD HOLDER NAME:

CARDHOLDER ZIP CODE

ORDER PLACED BY:

PHONE NO: ()

FAX NO: ()

SHIP TO

COMPANY NAME

ATTN:

ADDRESS

CITY

STATE

ZIP

BILL TO

COMPANY NAME

ATTN:

ADDRESS

CITY

STATE

ZIP

Technical Associates offers vibration diagnostic investigation services and support for all phases of your
vibration program from the initial startup through the analysis of the collected data.

R2005 07-21

TECHNICAL ASSOCIATES OF CHARLOTTE, P.C.

POST-SEMINAR EVALUATION

SEMINAR DATES: _____ SEMINAR LEVEL (CIRCLE): ENTRY I II III ADV.
 ATTENDEE NAME: _____ TITLE: _____
 COMPANY NAME: _____ YOUR VIBRATION EXPERIENCE: YEARS _____ MONTHS _____
 COMPANY LOCATION: _____ YOUR SPECTRAL ANAL. EXPER: YEARS _____ MONTHS _____
 SEMINAR LOCATION: _____
 INSTRUCTOR: _____

EVALUATION TOPICS RATING >	LOW HIGH									
	1	2	3	4	5	6	7	8	9	10
1. EVALUATE CONTENT OF THIS SEMINAR.										
2. EVALUATE PERFORMANCE OF INSTRUCTOR:										
A. KNOWLEDGE OF SUBJECT										
B. PRESENTATION CAPABILITIES										
C. ENTHUSIASM DISPLAYED										
D. OVERALL PERFORMANCE RATING										
3. HOW APPLICABLE TO YOUR JOB HAS BEEN THE CONTENT OF THIS SEMINAR AND SEMINAR TEXT?										
4. CAN WHAT YOU HAVE LEARNED MAKE YOU MORE EFFECTIVE IN YOUR JOB?										
5. HOW WOULD YOU RATE THE SEMINAR FACILITIES:										
A. MEETING ROOM -										
B. SLEEPING ROOM -										
C. FOOD SERVICE -										
6. ARE YOU INTERESTED IN FURTHER TRAINING IN THIS AREA OF VIBRATION/PREDICTIVE MAINTENANCE?										

CONCENTRATION ON SEMINAR:

- ON WHAT SHOULD MORE TIME BE CONCENTRATED? _____

- ON WHAT SHOULD LESS TIME BE CONCENTRATED? _____

- WHAT WOULD YOU SUGGEST TO IMPROVE FUTURE SEMINARS? _____

- DO YOU HAVE ANY ADDITIONAL COMMENTS ABOUT THIS SEMINAR, THE INSTRUCTORS, THE COMPANY OR THE FACILITIES? _____

WE WANT TO GENUINELY THANK YOU FOR THE PRIVILEGE OF SHARING WITH YOU THIS WEEK.
 WE HOPE TO SEE YOU AGAIN IN THE FUTURE.

Glossary

A

Acceleration. The time rate of change of velocity. Typical units are ft/s/s, meters/s/s, and G's ($1G = 32.17 \text{ ft/s/s} = 9.81 \text{ m/s/s}$). Acceleration measurements are usually made with accelerometers.

Accelerometer. Transducer whose output is directly proportional to acceleration. Most commonly use piezoelectric crystals to produce output.

Aliasing. A phenomenon which can occur whenever a signal is not sampled at greater than twice the maximum frequency component. Causes high frequency signals to appear at low frequencies. Aliasing is avoided by filtering out signals greater than $1/2$ the sample rate.

Alignment. A condition whereby the axes of machine components are either coincident, parallel or perpendicular, according to design requirements.

Amplification Factor (Synchronous). A measure of the susceptibility of a rotor to vibration amplitude when rotational speed is equal to the rotor natural frequency (implies a flexible rotor). For imbalance type excitation, synchronous amplification factor is calculated by dividing the amplitude value at the resonant peak by the amplitude value at a speed well above resonance (as determined from a plot of synchronous response vs. rpm).

Amplitude. The magnitude of dynamic motion or vibration. Amplitude is expressed in terms of peak-to-peak, zero-to-peak, or rms. For pure sine waves only, these are related as follows: rms = 0.707 times zero-to-peak; peak-to-peak = 2 times zero-to-peak. DSAs generally read rms for spectral components, and peak for time domain components.

Anti-Aliasing Filter. A low-pass filter designed to filter out frequencies higher than $1/2$ the sample rate in order to prevent aliasing.

Anti-Friction Bearing. See Rolling Element Bearing.

Asymmetrical Support. Rotor support system that does not provide uniform restraint in all radial directions. This is typical for most heavy industrial machinery where stiffness in one plane may be substantially different than stiffness in the perpendicular plane. Occurs in bearings by design, or from preloads such as gravity or misalignment.

Asynchronous. Vibration components that are not related to rotating speed (also referred to as nonsynchronous).

Attitude Angle (Steady-State). The angle between the direction of steady-state preload through the bearing centerline, and a line drawn between the shaft centerline and the bearing centerline. (Applies to fluid film bearings.)

Auto Spectrum (Power Spectrum). DSA spectrum display whose magnitude represents the power at each frequency, and which has no phase. Rms averaging produces an auto spectrum.

Averaging. In a DSA, digitally averaging several measurements to improve accuracy or to reduce the level of asynchronous components. Refer to definitions of rms, time, and peak-hold averaging.

Axial. In the same direction as the shaft centerline.

Axial Position. The average position, or change in position, of a rotor in the axial direction with respect to some fixed reference position. Ideally the reference is a known position within the thrust bearing axial clearance or float zone, and the measurement is made with a displacement transducer observing the thrust collar.

B

Balancing Resonance Speed(s). A rotative speed that corresponds to a natural resonance frequency.

Balanced Condition. For rotating machinery, a condition where the shaft geometric centerline coincides with the mass centerline.

Balancing. A procedure for adjusting the radial mass distribution of a rotor so that the mass centerline approaches the rotor geometric centerline.

Bandpass Filter. A filter with a single transmission band extending from lower to upper cutoff frequencies. The width of the band is determined by the separation of frequencies at which amplitude is attenuated by 3 dB (0.707).

Bandwidth. The spacing between frequencies at which a bandpass filter attenuates the signal by 3 dB. In a DSA, measurement bandwidth is equal to $[(\text{frequency span})/(\text{number of filters}) \times (\text{window factor})]$. Window factors are: 1 for uniform, 1.5 for Hanning, and 3.63 for Flat Top.

Baseline Spectrum. A vibration spectrum taken when a machine is in good operating condition; used as a reference for monitoring and analysis.

Blade Passing Frequency. A potential vibration frequency on any bladed machine (turbine, axial compressor, fan, etc.). It is represented by the number of blades times shaft-rotating frequency.

Block Size. The number of samples used in a DSA to compute the Fast Fourier Transform. Also the number of samples in a DSA time display. Most DSAs use a block size of 1024. Smaller block size reduces resolution.

Bode'. Rectangular coordinate plot of 1x component amplitude and phase (relative to a keyphasor) vs. running speed.

BPFO, BPFI. Common abbreviations for ball pass frequency of defects on outer and inner bearing races, respectively.

Bow. A shaft condition such that the geometric centerline of the shaft is not straight.

Brinelling (False). Impressions made by bearing rolling elements on the bearing race; typically caused by external vibration when the shaft is stationary.

C

Calibration. A test during which known values of the measured variable are applied to the transducer or readout instrument, and output readings varied or adjusted.

Campbell Diagram. A mathematically constructed diagram used to check for coincidence of vibration sources (i.e. 1x imbalance, 2x misalignment) with rotor natural resonances. The form of the diagram is a rectangular plot of resonant frequency (y-axis) vs excitation frequency (x-axis). Also known as an interference diagram.

Cascade Plot. See Spectral Map.

Cavitation. A condition which can occur in liquid handling machinery (e.g. centrifugal pumps) where a system pressure decrease in the suction line and pump inlet lowers fluid pressure and vaporization occurs. The result is mixed flow which may produce vibration.

Center Frequency. For a bandpass filter, the center of the transmission band.

Charge Amplifier. Amplifier used to convert accelerometer output impedance from high to low, making calibration much less dependent on cable capacitance.

Coherence. The ratio of coherent output power between channels in a dual-channel DSA. An effective means of determining the similarity of vibration at two locations, giving insight into the possibility of cause and effect relationships.

Constant Bandwidth Filter. A bandpass filter whose bandwidth is independent of center frequency. The filters simulated digitally in a DSA are constant band width.

Constant Percentage Bandwidth. A bandpass filter whose bandwidth is a constant percentage of center frequency. 1/3 octave filters, including those synthesized in DSAs, are constant percentage bandwidth.

Critical Machinery. Machines which are critical to a major part of the plant process. These machines are usually unspared.

Critical Speeds. In general, any rotating speed which is associated with high vibration amplitude. Often, the rotor speeds which correspond to natural frequencies of the system.

Critical Speed Map. A rectangular plot of system natural frequency (y-axis) versus bearing or support stiffness (x-axis).

Cross Axis Sensitivity. A measure of off-axis response of velocity and acceleration transducers.

Cycle. One complete sequence of values of a periodic quantity.

D

Damping. The quality of a mechanical system that restrains the amplitude of motion with each successive cycle. Damping of shaft motion is provided by oil in bearings, seals, etc. The damping process converts mechanical energy to other forms, usually heat.

Damping, Critical. The smallest amount of damping required to return the system to its equilibrium position without oscillation.

Decibels (dB). A logarithmic representation of amplitude ratio, defined as 20 times the base ten logarithm of the ratio of the measured amplitude to a reference. dBV readings, for example, are referenced to 1 volt rms. dB amplitude scales are required to display the full dynamic range of a DSA.

Degrees Of Freedom. A phrase used in mechanical vibration to describe the complexity of the system. The number of degrees of freedom is the number of independent variables describing the state of a vibrating system.

Digital Filter. A filter which acts on data after it has been sampled and digitized. Often used in DSAs to provide anti-aliasing protection after internal re-sampling.

Differentiation. Representation in terms of time rate of change. For example, differentiating velocity yields acceleration. In a DSA, differentiation is performed by multiplication by $j\omega$, where ω is frequency multiplied by 2π (Differentiation can also be used to convert displacement to velocity.)

Discrete Fourier Transform. A procedure for calculating discrete frequency components (filters or lines) from sampled time data. Since the frequency domain result is complex (i.e., real and imaginary components), the number of points is equal to half the number of samples.

Displacement. The change in distance or position of an object relative to a reference.

Displacement Transducer. A transducer whose output is proportional to the distance between it and the measured object (usually the shaft).

DSA. See Dynamic Signal Analyzer.

Dual Probe. A transducer set consisting of displacement and velocity transducers. Combines measurement of shaft motion relative to the displacement transducer with velocity of the displacement transducer to produce absolute motion of the shaft.

Dual Voting. Concept where two independent inputs are required before action (usually machine shutdown) is taken. Most often used with axial position measurements, where failure of a single transducer might lead to an unnecessary shutdown.

Dynamic Motion. Vibratory motion of a rotor system caused by mechanisms that are active only when the rotor is turning at speeds above slow roll speed.

Dynamic Signal Analyzer (DSA). Vibration analyzer that uses digital signal processing and the Fast Fourier Transform to display vibration frequency components. DSAs also display the time domain and phase spectrum, and can usually be interfaced to a computer.

E

Eccentricity, Mechanical. The variation of the outer diameter of a shaft surface when referenced to the true geometric centerline of the shaft. Out-of-roundness.

Eccentricity Ratio. The vector difference between the bearing centerline and the average steady-state journal centerline.

Eddy Current. Electrical current which is generated (and dissipated) in a conductive material in the presence of an electromagnetic field.

Electrical Runout. An error signal that occurs in eddy current displacement measurements when shaft surface conductivity varies.

Engineering Units. In a DSA, refers to units that are calibrated by the user (e.g., in/s, g's).

External Sampling. In a DSA, refers to control of data sampling by a multiplied tachometer signal. Provides a stationary display of vibration with changing speed.

F

Fast Fourier Transform (FFT). A computer (or microprocessor) procedure for calculating discrete frequency components from sampled time data. A special case of the discrete Fourier transform where the number of samples is constrained to a power of 2.

Filter. Electronic circuitry designed to pass or reject a specific frequency band.

Finite Element Modeling. A computer aided design technique for predicting the dynamic behavior of a mechanical system prior to construction. Modeling can be used, for example, to predict the natural frequencies of a flexible rotor.

Flat Top Filter. DSA window function which provides the best amplitude accuracy for measuring discrete frequency components.

Fluid-Film Bearing. A bearing which supports the shaft on a thin film of oil. The fluid-film layer may be generated by journal rotation (hydrodynamic bearing), or by externally applied pressure (hydrostatic bearing).

Forced Vibration. The oscillation of a system under the action of a forcing function. Typically forced vibration occurs at the frequency of the exciting force.

Free Vibration. Vibration of a mechanical system following an initial force-typically at one or more natural frequencies.

Frequency. The repetition rate of a periodic event, usually expressed in cycles per second (Hz), revolutions per minute (rpm), or multiples of a rotational speed (orders). Orders are commonly referred to as 1x for rotational speed, 2x for twice rotational speed, etc.

Frequency Response. The amplitude and phase response characteristics of a system.

G

G. The value of acceleration produced by the force of gravity.

Gear Mesh Frequency. A potential vibration frequency on any machine that contains gears; equal to the number of teeth multiplied by the rotational frequency of the gear.

H

Hanning Window. DSA window function that provides better frequency resolution than the flat top window, but with reduced amplitude accuracy.

Harmonic. Frequency component at a frequency that is an integer multiple of the fundamental frequency.

Heavy Spot. The angular location of the imbalance vector at a specific lateral location on a shaft. The heavy spot typically does not change with rotational speed.

Hertz (Hz). The unit of frequency represented by cycles per second.

High Spot. The angular location on the shaft directly under the vibration transducer at the point of closest proximity. The high spot can move with changes in shaft dynamics (e.g., from changes in speed).

High-Pass Filter. A filter with a transmission band starting at a lower cutoff frequency and extending to (theoretically) infinite frequency.

Hysteresis. Non-uniqueness in the relationship between two variables as a parameter increases or decreases. Also called deadband, or that portion of a system's response where a change in input does not produce a change in output.

I

Imbalance. Unequal radial weight distribution on a rotor system; a shaft condition such that the mass and shaft geometric centerlines do not coincide.

Impact Test. Response test where the broad frequency range produced by an impact is used as the stimulus. Sometimes referred to as a bump test.

Impedance, Mechanical. The mechanical properties of a machine system (mass, stiffness, damping) that determine the response to periodic forcing functions.

Influence Coefficients. Mathematical coefficients that describe the influence of system loading on system deflection.

Integration. A process producing a result that, when differentiated, yields the original quantity. Integration of acceleration, for example, yields velocity. Integration is performed in a DSA by dividing by $j\omega$, where ω is frequency multiplied by 2π . (Integration is also used to convert velocity to displacement).

J

Journal. Specific portions of the shaft surface from which rotor applied loads are transmitted to bearing supports.

K

Keyphasor. A signal used in rotating machinery measurements, generated by a transducer observing a once-per-revolution event. The keyphasor signal is used in phase measurements for analysis and balancing. (Keyphasor is a Bently Nevada trade name.)

L

Lateral Location. The definition of various points along the shaft axis of rotation.

Lateral Vibration. See Radial Vibration.

Leakage. In DSAs, a result of finite time record length that results in smearing of frequency components. Its effects are greatly reduced by the use of weighted window functions such as Flat Top and Hanning.

Linearity. The response characteristics of a linear system remain constant with input level. That is, if the response to input a is A, and the response to input b is B, then the response of a linear system to input (a + b) will be (A + B). An example of a non-linear system is one whose response is limited by mechanical stop, such as occurs when a bearing mount is loose.

Lines. Common term used to describe the filters of a DSA (e.g., 400 line analyzer).

Linear Averaging. See Time Averaging.

Low-Pass Filter. A filter whose transmission band extends from dc to an upper cutoff frequency.

M

Mechanical Runout. An error in measuring the position of the shaft centerline with a displacement probe that is caused by out-of-roundness and surface imperfections.

Micrometer (MICRON). One millionth (.000001) of a meter. (1 micron = 1×10^{-6} meters = 0.04 mils.)

MIL. One thousandth (0.001) of an inch. (1 mil = 25.4 microns.)

Modal Analysis. The process of breaking complex vibration into its component modes of vibration, very much like frequency domain analysis breaks vibration down to component frequencies.

Mode Shape. The resultant deflected shape of a rotor at a specific rotational speed to an applied forcing function. A three-dimensional presentation of rotor lateral deflection along the shaft axis.

Modulation, Amplitude (AM). The process where the amplitude of a signal is varied as a function of the instantaneous value of another signal. The first signal is called the carrier, and the second signal is called the modulating signal. Amplitude modulation produces a component at the carrier frequency, with adjacent components (sidebands) at the frequency of the modulating signal.

Modulation, Frequency (FM). The process where the frequency of the carrier is determined by the amplitude of the modulating signal. Frequency modulation produces a component at the carrier frequency, with adjacent components (sidebands) at the frequency of the modulating signal.

N

Natural Frequency. The frequency of free vibration of a system. The frequency at which an undamped system with a single degree of freedom will oscillate upon momentary displacement from its rest position.

Nodal Point. A point of minimum shaft deflection in a specific mode shape. May readily change location along the shaft axis due to changes in residual imbalance or other forcing function, or change in restraint such as increased bearing clearance.

Noise. Any component of a transducer output signal that does not represent the variable intended to be measured.

Nyquist Criterion. Requirement that a sampled system sample at a frequency greater than twice the highest frequency to be measured.

Nyquist Plot. A plot of real versus imaginary spectral components that is often used in servo analysis. Should not be confused with a polar plot of amplitude and phase of 1x vibration.

O

Octave. The interval between two frequencies with a ratio of 2 to 1.

Oil Whirl/Whip. An unstable free vibration whereby a fluid-film bearing has insufficient unit loading. Under this condition, the shaft centerline dynamic motion is usually circular in the direction of rotation. Oil whirl occurs at the oil flow velocity within the bearing, usually 40 to 49% of shaft speed. Oil whip occurs when the whirl frequency coincide with (and becomes locked to) a shaft resonant frequency. (Oil whirl and whip can occur in any case where fluid is between two cylindrical surfaces.)

Orbit. The path of the shaft centerline motion during rotation. The orbit is observed with an oscilloscope connected to x and y-axis displacement transducers. Some dual-channel DSAs also have the ability to display orbits.

Oscillator-Demodulator. A signal conditioning device that sends a radio frequency signal to an eddy-current displacement probe, demodulates the probe output, and provides output signals proportional to both the average and dynamic gap distances. (Also referred to as Proximitors, a Bently Nevada trade name.)

P

Peak Hold. In a DSA, a type of averaging that holds the peak signal level for each frequency component.

Period. The time required for a complete oscillation or for a single cycle of events. The reciprocal of frequency.

Phase. A measurement of the timing relationship between two signals, or between a specific vibration event and a keyphasor pulse.

Piezoelectric. Any material which provides a conversion between mechanical and electrical energy. For a piezoelectric crystal, if mechanical stresses are applied on two opposite faces, electrical charges appear on some other pair of faces.

Polar Plot. Polar coordinate representation of the locus of the 1x vector at a specific lateral shaft location with the shaft rotational speed as a parameter.

Power Spectrum. See Auto Spectrum.

Preload, Bearing. The dimensionless quantity that is typically expressed as a number from zero to one where a preload of zero indicates no bearing load upon the shaft, and one indicates the maximum preload (i.e., line contact between shaft and bearing).

Preload, External. Any of several mechanisms that can externally load a bearing. This includes 'soft' preloads such as process fluids or gravitational forces as well as "hard" preloads from gear contact forces, misalignment, rubs, etc.

R

Radial. Direction perpendicular to the shaft centerline.

Radial Position. The average location, relative to the radial bearing centerline, of the shaft dynamic motion.

Radial Vibration. Shaft dynamic motion or casing vibration which is in a direction perpendicular to the shaft centerline.

Real-Time Analyzer. See Dynamic Signal Analyzer.

Real-Time Rate. For a DSA, the broadest frequency span at which data is sampled continuously. Real-time rate is mostly dependent on FFT processing speed.

Rectangular Window. See Uniform Window.

Relative Motion. Vibration measured relative to a chosen reference. Displacement transducers generally measure shaft motion relative to the transducer mounting.

Repeatability. The ability of a transducer or readout instrument to reproduce readings when the same input is applied repeatedly.

Resolution. The smallest change in stimulus that will produce a detectable change in the instrument output.

Resonance. The condition of vibration amplitude and phase change response caused by a corresponding system sensitivity to a particular forcing frequency. A resonance is typically identified by a substantial amplitude increase, and related phase shift.

Rolling Element Bearing. Bearing whose low friction qualities derive from rolling elements (balls or rollers), with little lubrication.

Root Mean Square (rms). Square root of the arithmetical average of a set of squared instantaneous values. DSAs perform rms averaging digitally on successive vibration spectra.

Rotor, Flexible. A rotor which operates close enough to, or beyond its first bending critical speed for dynamic effects to influence rotor deformations. Rotors which cannot be classified as rigid rotors are considered to be flexible rotors.

Rotor, Rigid. A rotor which operates substantially below its first bending critical speed. A rigid rotor can be brought into, and will remain in, a state of satisfactory balance at all operating speeds when balanced on any two arbitrarily selected correction planes.

RPM Spectral Map. A spectral map of vibration spectra versus rpm.

Runout Compensation. Electronic correction of a transducer output signal for the error resulting from slow roll runout.

S

Seismic. Refers to an inertially referenced measurement or a measurement relative to free space.

Seismic Transducer. A transducer that is mounted on the case or housing of a machine and measures casing vibration relative to free space. Accelerometers and velocity transducers are seismic.

Signal Conditioner. A device placed between a signal source and a readout instrument to change the signal. Examples: attenuators, preamplifiers, charge amplifiers.

Signature. Term usually applied to the vibration frequency spectrum which is distinctive and special to a machine or component, system or subsystem at a specific point in time, under specific machine operating conditions, etc. Used for historical comparison of mechanical condition over the operating life of the machine.

Slow Roll Speed. Low rotative speed at which dynamic motion effects from forces such as imbalance are negligible.

Spectral Map. A three-dimensional plot of the vibration amplitude spectrum versus another variable, usually time or rpm.

Spectrum Analyzer. An instrument which displays the frequency spectrum of an input signal.

Stiffness. The spring-like quality of mechanical and hydraulic elements to elasticity deform under load.

Strain. The physical deformation, deflection, or change in length resulting from stress (force per unit area).

Subharmonic. Sinusoidal quantity of a frequency that is an integral submultiple of a fundamental frequency.

Subsynchronous. Component(s) of a vibration signal which has a frequency less than shaft rotative frequency.

Synchronous Sampling. In a DSA, it refers to the control of the effective sampling rate of data; which includes the processes of external sampling and computed resampling used in order tracking.

T

Time Averaging. In a DSA, averaging of time records that results in reduction of asynchronous components.

Time Record. In a DSA, the sampled time data converted to the frequency domain by the FFT. Most DSAs use a time record of 1024 samples.

Torsional Vibration. Amplitude modulation of torque measured in degrees peak-to-peak referenced to the axis of shaft rotation.

Tracking Filter. A low-pass or bandpass filter which automatically tracks the input signal. A tracking filter is usually required for aliasing protection when data sampling is controlled externally.

Transducer. A device for translating the magnitude of one quantity into another quantity.

Transient Vibration. Temporarily sustained vibration of a mechanical system. It may consist of forced or free vibration or both. Typically this is associated with changes in machine operating condition such as speed, load, etc.

Transverse Sensitivity. See Cross-Axis Sensitivity.

Trigger. Any event which can be used as a timing reference. In a DSA, a trigger can be used to initiate a measurement.

U

Unbalance. See Imbalance.

Uniform Window. In a DSA, a window function with uniform weighting across the time record. This window does not protect against leakage, and should be used only with transient signals contained completely within the time record.

V

Vector. A quantity which has both magnitude and direction (phase).

W

Waterfall Plot. See Spectral Map.